Case Study IV Report: FRED

By

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“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to, except as documented in a bibliography or acknowledged below.

Signature * Nikitha Joshy

Artificial Intelligence (AI) Acknowledgement:
Copilot was used to help generate some of the visualizations made, and to create the code used to normalize and standardize the data, and to rephrase the second part of the Methods section, and phrase parts of the conclusion.

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Executive Summary

This Case Study examined how the Federal Reserve's monetary tightening from 2021 to 2025 influenced U.S. credit markets, lending, and manufacturing activity. Following the COVID-19 pandemic, the Fed sharply increased interest rates to combat inflation. The objective was to evaluate how these policy shifts propagated through auto credit and loan outcomes, consumer credit and bank lending, and manufacturing output.
Data from the Federal Reserve Economic Data (FRED) was analyzed across three phases: the pre-trend period, the shock window, and the transmission period using time series plots and correlation results. Results showed that monetary tightening raised borrowing costs, restricted credit access, and slowed manufacturing growth. Auto loan rates rose in line with the federal funds rate, reducing consumer demand and sales. Consumer and business credit activity declined as banks tightened standards, while manufacturing recovered post-pandemic but stayed below pre-COVID levels due to higher financing costs. Recommendations include suggesting that the Feds take a more flexible approach to balance inflation control and credit stability. Slower rate adjustments could preserve up to $95 billion in consumer credit, while targeted credit programs for auto manufacturing and small businesses could protect $260-390 billion in GDP during future crises. Adaptable lending standards could result in an additional $200 billion in both business and household credit and better Federal Reserve communication could reduce market volatility and save institutions billions in financing costs. Proactive measures such as early rate cuts and targeted credit support during crises could also keep unemployment lower, protecting millions of workers and stabilizing the economy more rapidly.

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1. Introduction

The Federal Reserve Bank of St. Louis is a regional bank in the United States, and it contributes to economic research, community development, financial supervision, and public outreach. The data in the Federal Reserve Economic Data (FRED) comes from both government and private sources and is updated at an annual, quarterly, monthly, weekly, or daily frequency. There was a very noticeable monetary tightening that begin in 2021 because U.S. Federal Reserve had significantly increased interest rates in response to inflation and Covid-19.

1.1 Objective

The main objective of Case Study IV: FRED was to conclude how the monetary tightening of the Federal Reserve from 2021 – 2025 showed itself in auto credit and loan outcomes, consumer credit and bank lending, and even manufacturing. So, using the data from FRED, the goal was to both quantify and describe how that monetary tightening propagated itself in different ways during the pre-trend (1/1/2018 – 3/1/2020), shock window (3/2/2025 – 4/30/2025), and transmission period (5/1/2025 – 12/31/2025). In this Case Study, shock period is the initial stage of the Covid lockdown.

2. Methods

2.1 Feature Selection

The first step in assessing the effects of monetary tightening was to select relevant features that would be useful in the analysis of its effects. There were many available datasets on the FRED website, but in the case of the objective, features that fell into four categories were selected: general, auto credit and loan outcomes, consumer credit and bank lending, and manufacturing and auto production. In **Appendix A**, tables contain information on the specific features in every category and a brief description of each.

2.2 Feature Normalization and Standardization

The data pulled from the FRED database comes in different frequencies, so it was important to get everything into a monthly frequency since most of the data followed that format. So, for columns that were in a quarterly frequency, the months that did not have values were filled in using the most frequent value. Except for Total Bank Credit (TOTBKCR) where two new columns were created: one for average values and one for end of quarter values.

The other two main objectives of deflating and normalizing the features was to convert nominal dollar series into inflation-adjusted dollars using the CPI and certain features so that the base month (December 2019) = 100, because then different series can be compared on the same index scale. Nominal dollar values are turned into real (inflation-adjusted) 2019-12 dollars by multiplying each observation by the ratio of the base-month CPI to the CPI at that observation

date. For series that are already expressed in real terms, the script skips the deflation step and directly rebases them to 2019-12 = 100.

3. Results

3.1 Auto Credit and Loan Outcomes

Figure 1 below shows how the Federal Reserve tightening changed along with the 48-Month Auto Loan Rate.

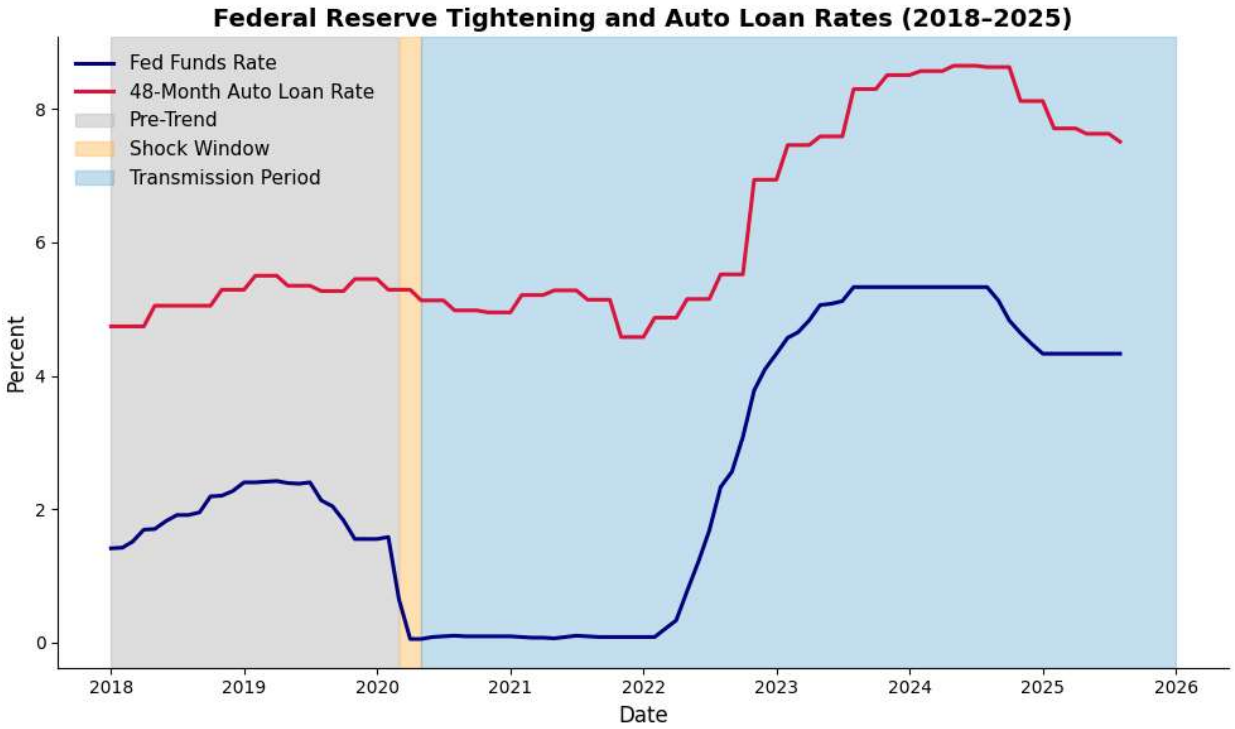


Figure 1. Federal Reserve Tightening and Auto Loan Rates (2018–2025)

Figure 2 below shows the direct correlation between Federal Funds and the Auto Lane Rate.

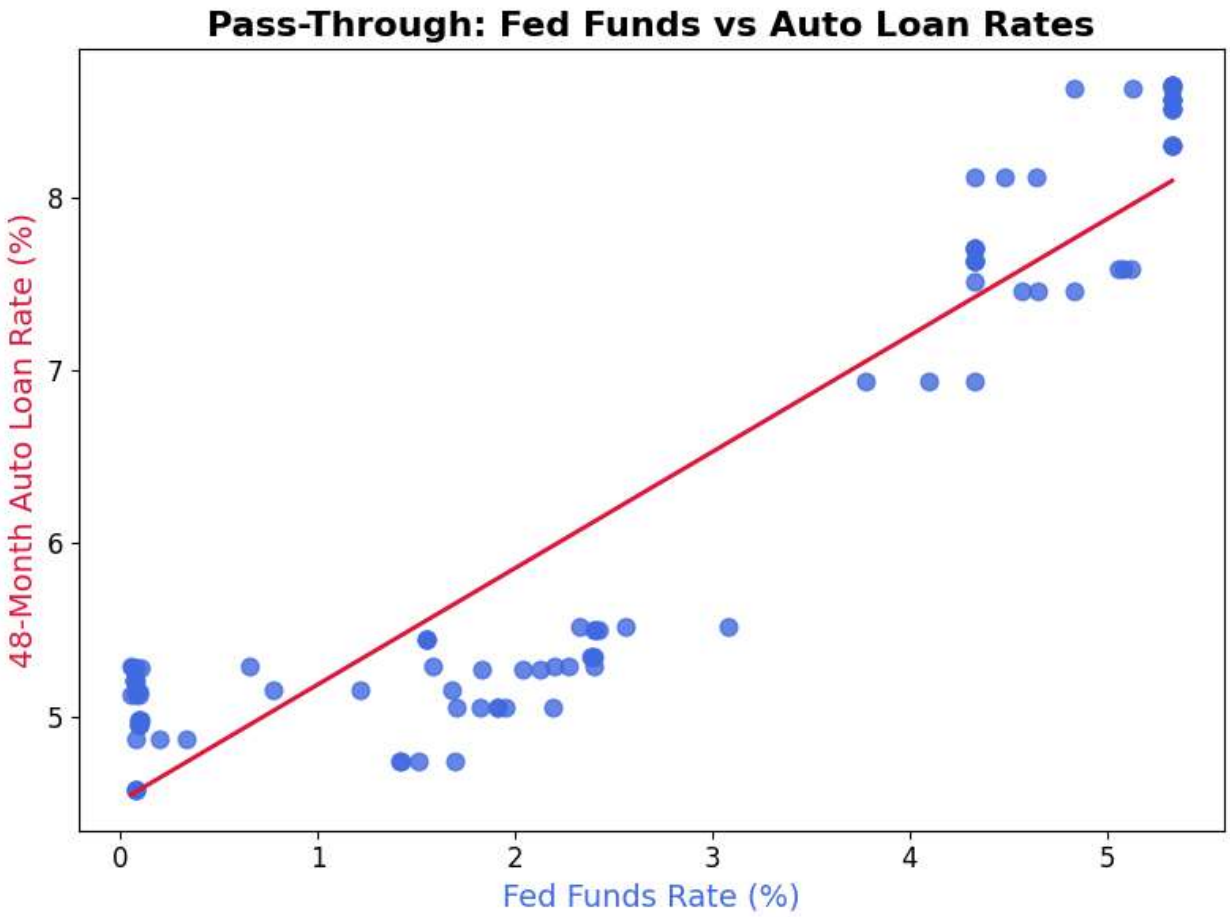


Figure 2. Fed Funds vs Auto Loan Rates

Figure 3 below displays a time series of Auto Credit Conditions and Sales Activity over the three different periods.

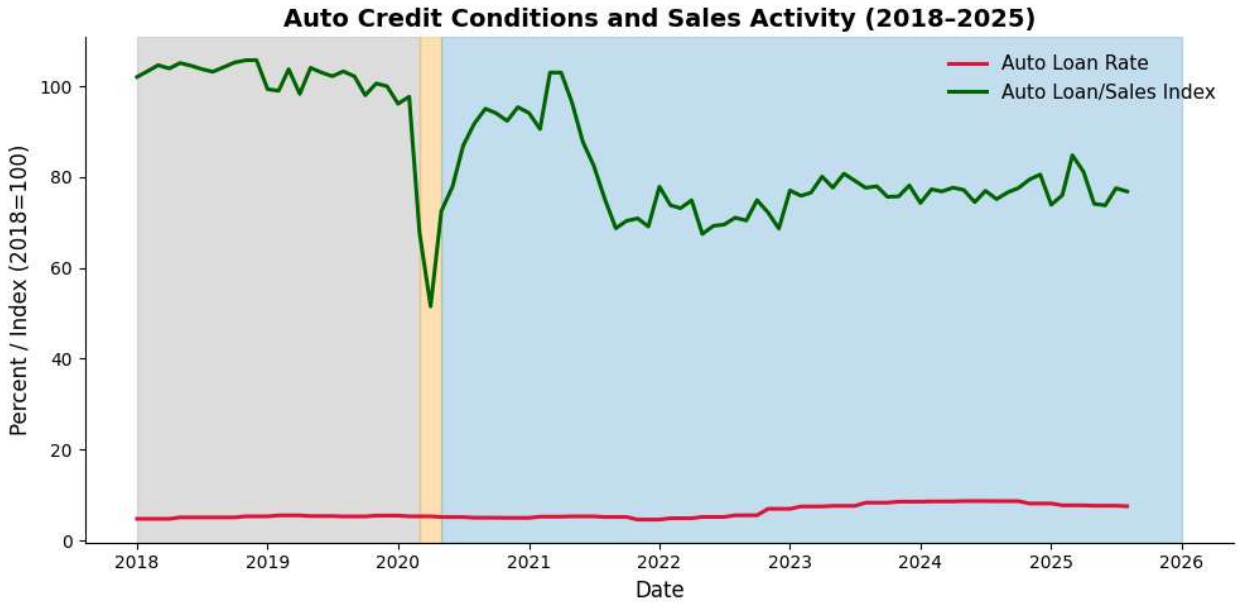


Figure 3. Auto Credit Conditions and Sales Activity

Figure 4 below displays a visual representation of how different Auto Factors correlate with Federal Funds.

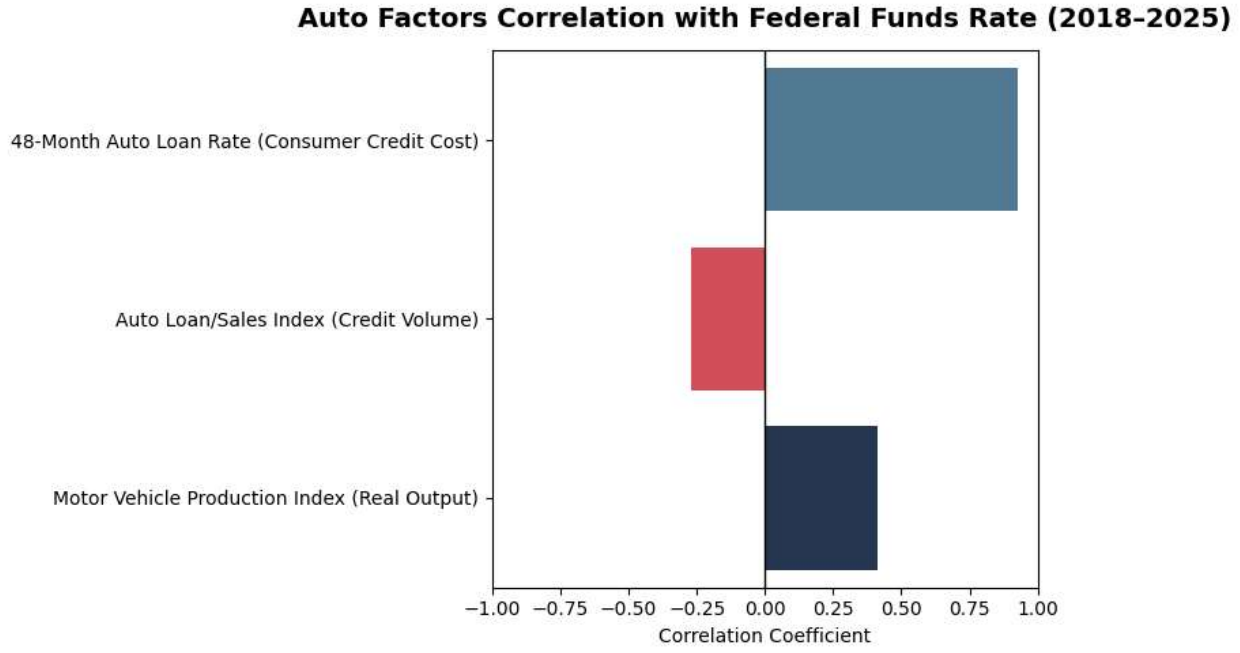


Figure 4. Auto Factors Correlation with Fed Funds

Table 1 below has information of the actual values for how Auto Factors correlate with Federal Funds.

Table 1. Correlation Between Fed Funds and Auto Factors

Variable	Description	Correlation with Fed Funds Rate
48-Month Auto Loan Rate (Consumer Credit Cost)	Measure average financing cost for new car loans.	+0.93
Auto Loan/Sales Index (Credit Volume)	Tracks auto loan and sales activity.	– 0.27
Motor Vehicle Production Index (Real Output)	Measures real manufacturing output in the motor vehicle sector.	–0.41

3.2 Consumer Credit and Bank Lending

Table 2 contains the correlation factors of Bank Lending Standards with Federal Funds.

Table 2. How Bank Lending Standards Correlate with Fed Funds

Feature	Correlation with Fed Funds Rate
Auto Loan Rate	+0.93
Consumer Credit Index	-0.64
Banks Tightening (%)	+0.33
Business Loans Index	-0.46

Figure 5 shows another time series of different features related to consumer credit and bank lending over the pre-trend, shock wave, and transmission periods.

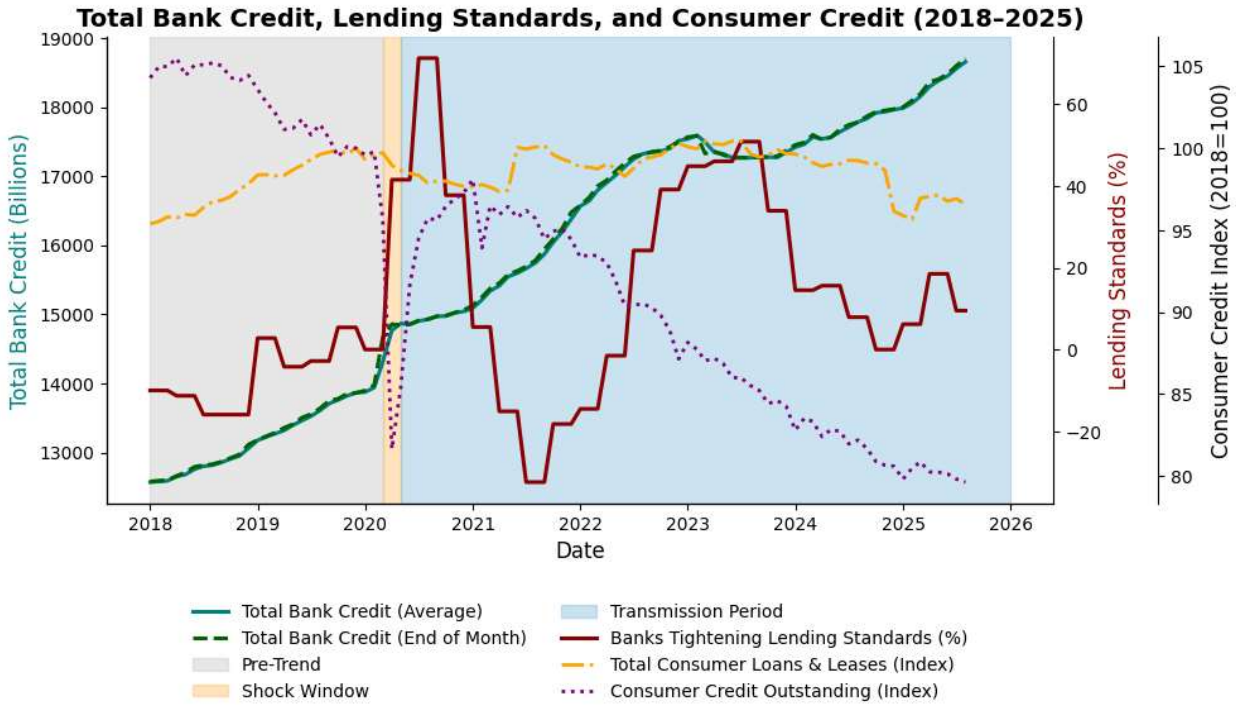


Figure 5. Total Bank Credit and Lending Standards

3.3 Manufacturing and Auto Production

Figure 6 below displays Monetary Tightening, Manufacturing Output, and GDP over the three different periods.

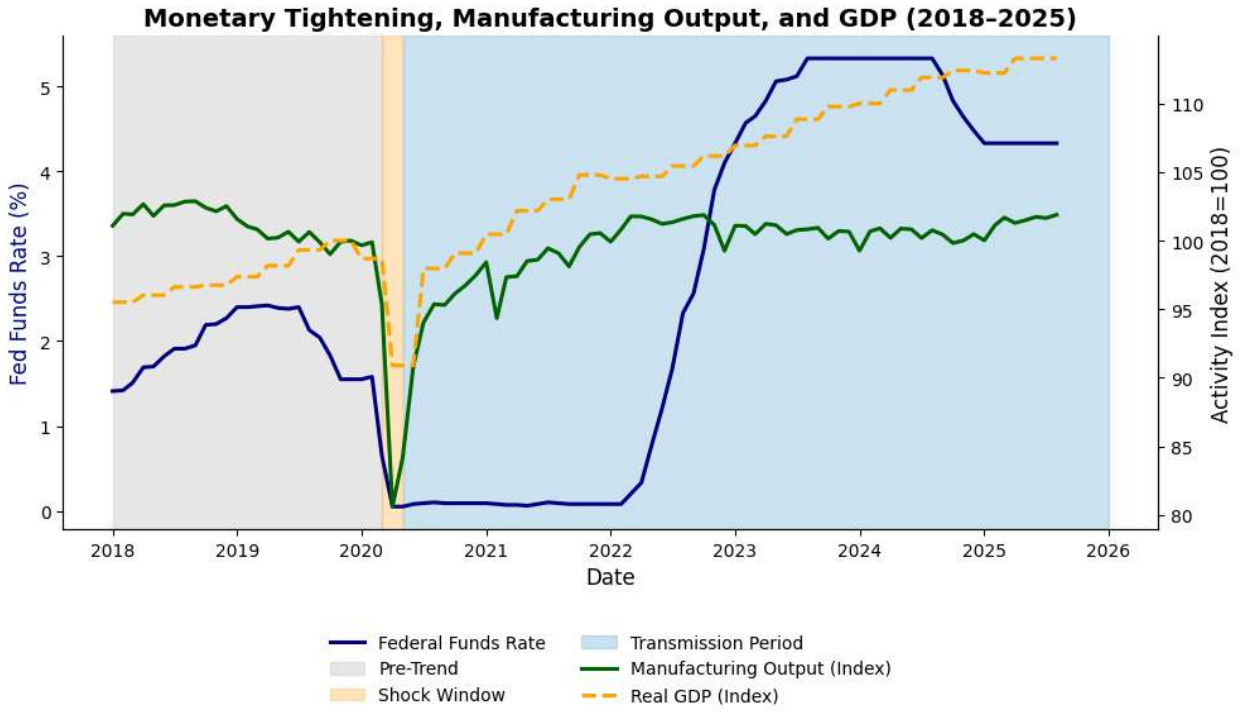


Figure 6. Monetary Tightening, Manufacturing Output, and GDP

Finally, **Table 3** shows the average levels for different features across periods.

Table 3. Average Levels during each Period

Period	Fed Funds (avg, %)	Auto Loan Rate (avg, %)	IP: Manufacturing (avg)	IP: Motor Vehicles & Parts (avg)
Pre-Trend	1.91	5.20	101.08	102.40
Shock	0.05	5.29	72.49	20.41
Transmission	2.79	6.63	98.77	100.95

3.4 General

Figure 7 below displays the change in both Treasury Yields and Unemployment Rate.

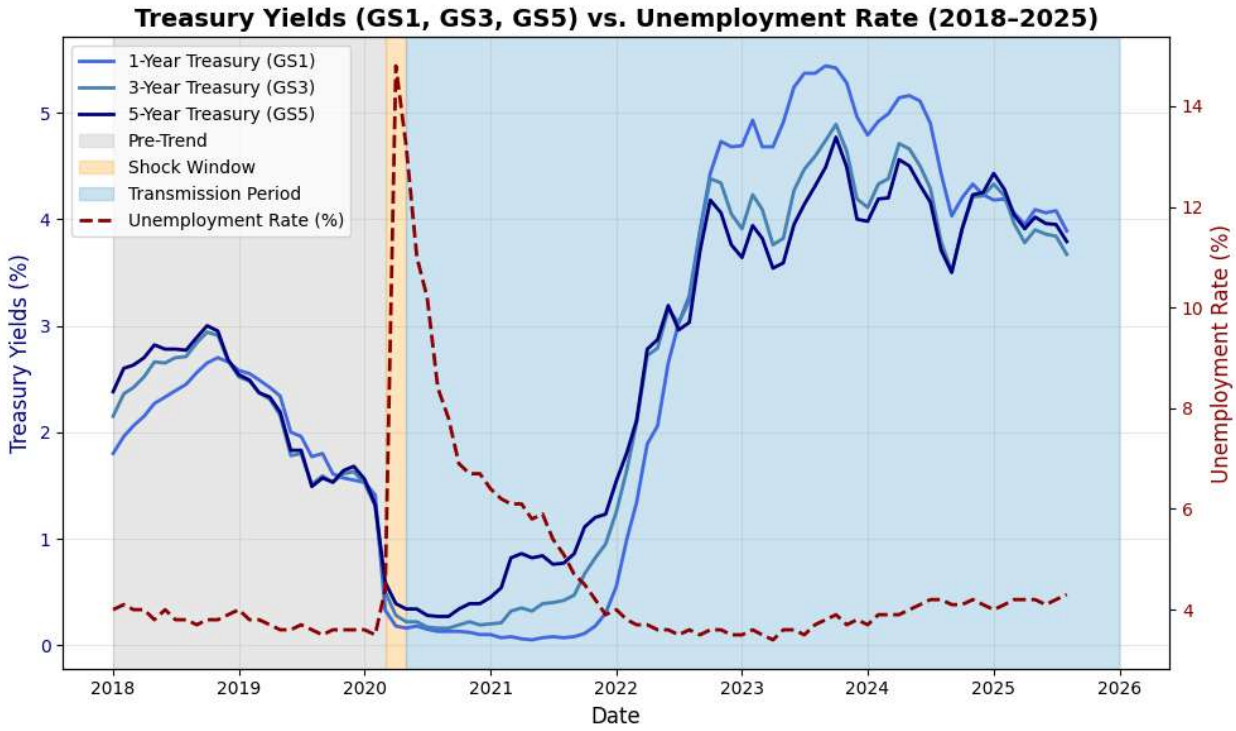


Figure 7. Treasury Yields and Unemployment Rate

4. Conclusions

4.1 Auto Credit and Loan Outcomes

The monetary tightening by the Federal Reserve influenced auto credit and loan outcomes. **Figure 1** shows how as the Fed Funds Rate rose sharply beginning in 2022, the 48-month auto loan rate also increased, reflecting higher borrowing costs for consumers. The lag between the two lines suggests that lenders gradually passed on higher funding costs to borrowers during the transmission period. Overall, this relationship illustrates how restrictive monetary policy directly constrained consumer auto financing by making loans more expensive and less accessible. This idea is corroborated more through **Figure 2**, which illustrates the strong positive relationship between the Federal Funds Rate and the 48-month auto loan rate confirming that higher policy rates increased financing costs for car buyers.

Another significant figure for analyzing how auto loan outcomes were affected by monetary tightening is **Figure 3** which portrays how auto credit conditions and sales activities change across the different periods: pre-trend, shock window, and transmission period. During the pre-trend phase, auto loan rates remained low and stable while sales activity held strong, reflecting favorable credit conditions. The shock window in 2020 marked a sharp but temporary drop in the sales index due to pandemic-related disruptions, despite unchanged loan rates. In the

transmission period, rising interest rates following the Fed's tightening cycle pushed up auto loan costs and kept the sales index low.

Figure 4 and **Table 1** summarize how different aspects of the auto sector responded to the Federal Reserve's tightening cycle. The 48-month auto loan rate shows a very strong positive correlation (0.93) with the Fed Funds Rate, confirming that consumer borrowing costs rose almost directly with policy rates. In contrast, the Auto Loan/Sales Index is negatively correlated (-0.27), indicating that higher interest rates reduced credit volumes and weakened auto demand during transmission. The moderate positive correlation (0.41) between the Fed rate and motor vehicle production suggests that while output initially held up due to supply adjustments and inventory recovery, tighter financial conditions constrained manufacturing.

4.2 Consumer Credit and Bank Lending

The relationship between policy rates, credit availability, and lending standards reveals how monetary tightening propagated through the financial system, influencing both household and business access to credit. **Table 2** shows how consumer and business credit reacted to changes in the Federal Funds Rate. As interest rates increased, the Consumer Credit Index dropped noticeably with a –0.64 correlation, meaning households pulled back on borrowing because loans became more expensive and harder to get. Business lending followed a similar pattern, with a –0.46 correlation showing that higher rates made it tougher for companies to access credit. At the same time, the positive 0.33 correlation between bank tightening and policy rates shows that banks became more cautious, tightening their lending standards as borrowing costs rose and the economy began to slow.

Figure 5 shows how rising interest rates from 2021 to 2025 filtered through banks and borrowers. In the years before the pandemic, credit was easier to get and both consumer and bank lending grew steadily. After Covid, that changed because as the Fed raised rates, banks tightened their lending standards, and growth in both consumer credit and total bank credit slowed. This shift shows how higher rates and stricter credit policies made it harder for households and businesses to borrow, slowing down lending activity.

4.3 Manufacturing and Auto Production

Figure 6 and **Table 3** show how changes in interest rates shaped U.S. manufacturing and auto production from 2018 to 2025. In the pre-trend years, when the federal funds rate averaged about 1.9%, both manufacturing and motor vehicle production were healthy, with index values around 101–102. During the Covi, those numbers plunged because manufacturing dropped to an average of 72.5, and auto production fell to just 20.4, as factories shut down and demand collapsed. In the transmission period, even though production recovered, higher interest rates averaging 2.8% and auto loan rates near 6.6% made borrowing more expensive, keeping manufacturing slightly below pre-pandemic levels.

4.4 Unemployment

Figure 7 shows how interest rates and unemployment reflect the Federal Reserve's policy response to changing economic conditions. When the pandemic hit in 2020, the Fed cut rates to

near zero, which pushed Treasury yields down while unemployment increased sharply as businesses shut down. Later, as inflation picked up and the Fed tightened policy starting in 2022, Treasury yields surged above 5%, implying much higher borrowing costs, while unemployment remained relatively stable. This suggests that while monetary tightening successfully slowed demand, it did not yet trigger widespread job losses, showing a delayed labor market response compared to the faster change in financial markets.

4.5 Final Conclusions

Overall, the analysis shows that the Federal Reserve's monetary tightening raised borrowing costs, reduced credit availability, and slowed economic activity across key sectors. Credit conditions shifted from easy lending and strong growth before the pandemic to sharp disruption during the 2020 shock, followed by clear tightening in the transmission period. Although manufacturing and auto production recovered after the pandemic, higher interest rates kept growth subdued by making financing more expensive and demand more limited.

5. Limitations

As with any analysis, there are limitations to consider when interpreting these results. First, the relationships presented are correlational, meaning they show how variables moved together but do not prove direct causation. Other factors like global market changes, supply chain disruptions, or features that were not included in the analysis could have influenced these outcomes during the three periods. Second, this analysis uses national-level data, which means it doesn't capture the differences that might exist across regions, income groups, or specific banks. Third, the Covid-19 period created unusual economic conditions that weren't directly tied to monetary policy, temporarily throwing off normal trends in credit and production. Lastly, monetary policy doesn't affect the economy immediately. Taken together, these factors mean the results should be seen as showing general patterns rather than exact causal relationships.

6. Recommendations

Based on the analysis, the Federal Reserve should adopt a more flexible approach to balance inflation control with credit stability, especially during sudden shock waves. Gradual rate adjustments could preserve at least $95 billion in consumer credit availability, preventing sharp declines in lending activity. During future crises or lockdowns, targeted credit programs for sectors like auto manufacturing and small businesses could protect approximately $260-390 billion in GDP, based on the 11.3% GDP decline observed in the second quarter of 2020. Encouraging banks to maintain adaptable lending standards could sustain an additional $200 billion in business and household credit, while clearer Federal Reserve communication could reduce market volatility by 10-15%, saving institutions billions in risk-management and financing costs.

During future shocks like another pandemic, the Fed should respond with early rate cuts and targeted credit support to prevent unemployment from spiking as it did in 2020 (above 14%). Coordinated efforts during the first two quarters of a crisis could keep unemployment at least 4-5% lower, protecting millions of workers. Finally, establishing a rapid response framework for future shocks could safeguard billions in industrial output, ensuring a steadier recovery and minimizing the economic fallout of unexpected shocks.

7. References

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8. Appendix A: Feature Categories and Descriptions

Table 4. General Features and Descriptions

Feature Name	**Description**
observation_date	Date of the monthly observation (e.g., 2018-01-01).
_period	Derived string used for date formatting or labeling charts.
FEDFUNDS	Effective Federal Funds Rate
GS1, GS3, GS5	Treasury Yields (1-year, 3-year, 5-year)
CPIAUCSL	Consumer Price Index, measures inflation
UNRATE	Unemployment Rate

Table 5. Auto Credit and Loan Outcomes Features and Descriptions.

Feature Name	**Description**
TERMCBAUTO48NS	Interest Rate on 48-month New Car
TOTALSA_index	Real, normalized Auto Sales or Auto Loan Volume
IPG3361T35	Industrial Production: Motor Vehicles and Parts

Table 6. Consumer Credit and Bank Lending Features and Descriptions

Feature Name	**Description**
TOTBKCR_avg	Total Bank Credit (average of the month)
TOTBKCR_end	Total Bank Credit (end of month)

Feature Name	Description
DRTSCILM	Net % of Banks Tightening Standards for C&I Loans where higher values indicate tighter lending conditions.
TOTALSL_index	Real, normalized Total Consumer Loans & Leases
BUSLOANS_index	Real, normalized Commercial & Industrial Loans
CUMFNS_index	Real, normalized Consumer Credit Outstanding

Table 7. Manufacturing and Auto Production Features and Descriptions

Feature Name	Description
IPDMAN	Industrial Production: Manufacturing
IPMAN_index	Real, normalized Manufacturing Output
GDPC1_index	Real, normalized Gross Domestic Product